

HDOC DAY -23

QUESTION:

Write a C program to display a menu and print different patterns using a switch statement and nested loops without using arrays. Each pattern must be implemented in a separate parameterless function with no return value.

Aim

To develop a menu-driven C program using a switch statement and nested loops to display solid square, solid rectangular, row-number, and column-number patterns without using arrays.

Algorithm

1. Start.
2. Display the menu:
 1. Solid square pattern
 2. Solid rectangular pattern
 3. Row-number pattern
 4. Column-number pattern
3. Read the user's choice.
4. Use a switch statement to call the corresponding function.
5. Use nested loops inside the selected function to print the required pattern.
6. Stop.

Algorithm 1: Solid Square Pattern

1. Start.
2. Input the number of rows n.
3. Set i = 1.
4. Repeat while i <= n:
 - Set j = 1.
 - Print * while j <= n.
 - Move to the next line.
5. Stop.

Algorithm 2: Solid Rectangular Pattern

1. Start.
2. Input number of rows m and columns n .
3. Set $i = 1$.
4. Repeat while $i \leq m$:
 - Set $j = 1$.
 - Print $*$ while $j \leq n$.
 - Move to the next line.
5. Stop

Algorithm 3: Row-Number Pattern

1. Start.
2. Input number of rows m and columns n .
3. Set $i = 1$.
4. Repeat while $i \leq m$:
 - Set $j = 1$.
 - Print the row number i while $j \leq n$.
 - Move to the next line.
5. Stop.

Algorithm 4: Column-Number Pattern

1. Start.
2. Input number of rows m and columns n .
3. Set $i = 1$.
4. Repeat while $i \leq m$:
 - Set $j = 1$.
 - Print the column number j while $j \leq n$.
 - Move to the next line.
5. Stop.

Test-Case Table 1: Solid Square Pattern

Test Case Input n Expected Output

1	1	*
2	3	3 × 3 square of *
3	4	4 × 4 square of *

For Test Case 3, expected output:

```
* * * *
* * * *
* * * *
* * * *
```

Test-Case Table 2: Solid Rectangular Pattern

Test Case Input m (Rows) Input n (Columns) Expected Output

1	1	3	1 × 3 rectangle
2	2	4	2 × 4 rectangle
3	3	5	3 × 5 rectangle

For Test Case 2, expected output:

```
* * * *
* * * *
```

Test-Case Table 3: Row-Number Pattern

Test Case Input m (Rows) Input n (Columns) Expected Output

1	2	3	2 × 3 row-number pattern
2	3	4	3 × 4 row-number pattern
3	4	4	4 × 4 row-number pattern

For Test Case 3:

```
1 1 1 1
2 2 2 2
3 3 3 3
```

4 4 4 4

Test-Case Table 4: Column-Number Pattern

Test Case Rows m Columns n Expected Output

1	2	3	2 × 3 column-number pattern
2	3	4	3 × 4 column-number pattern
3	4	4	4 × 4 column-number pattern

For Test Case 3:

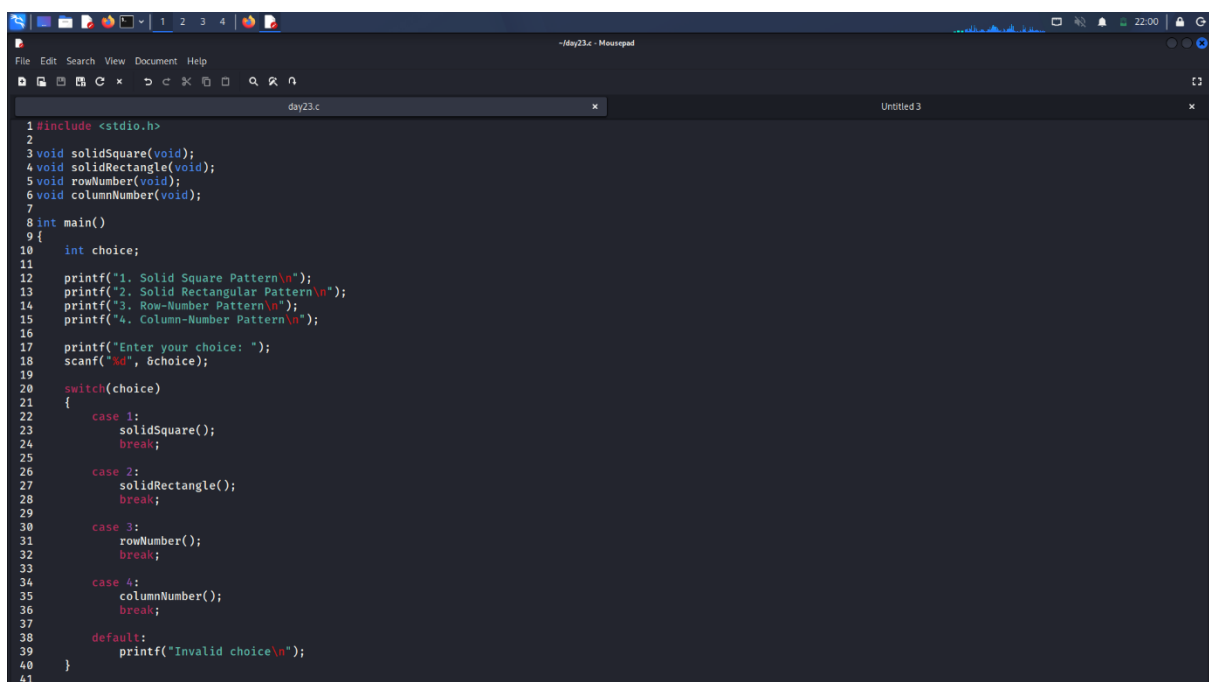
1 2 3 4

1 2 3 4

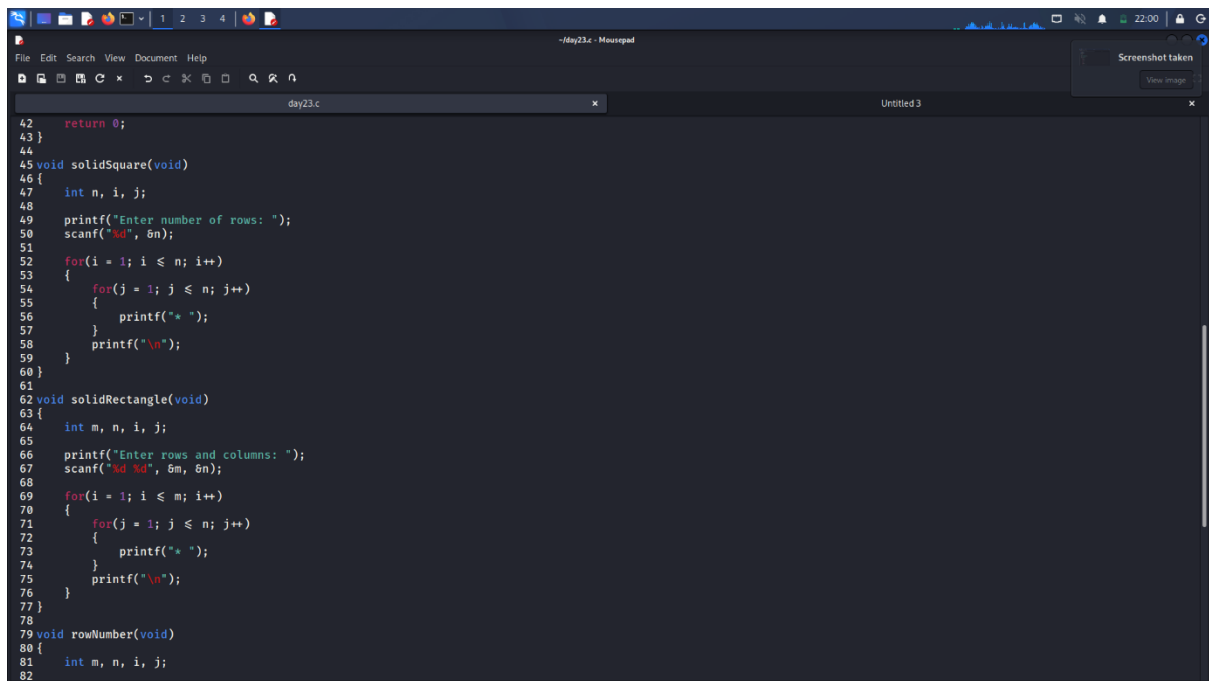
1 2 3 4

1 2 3 4

>C Program:

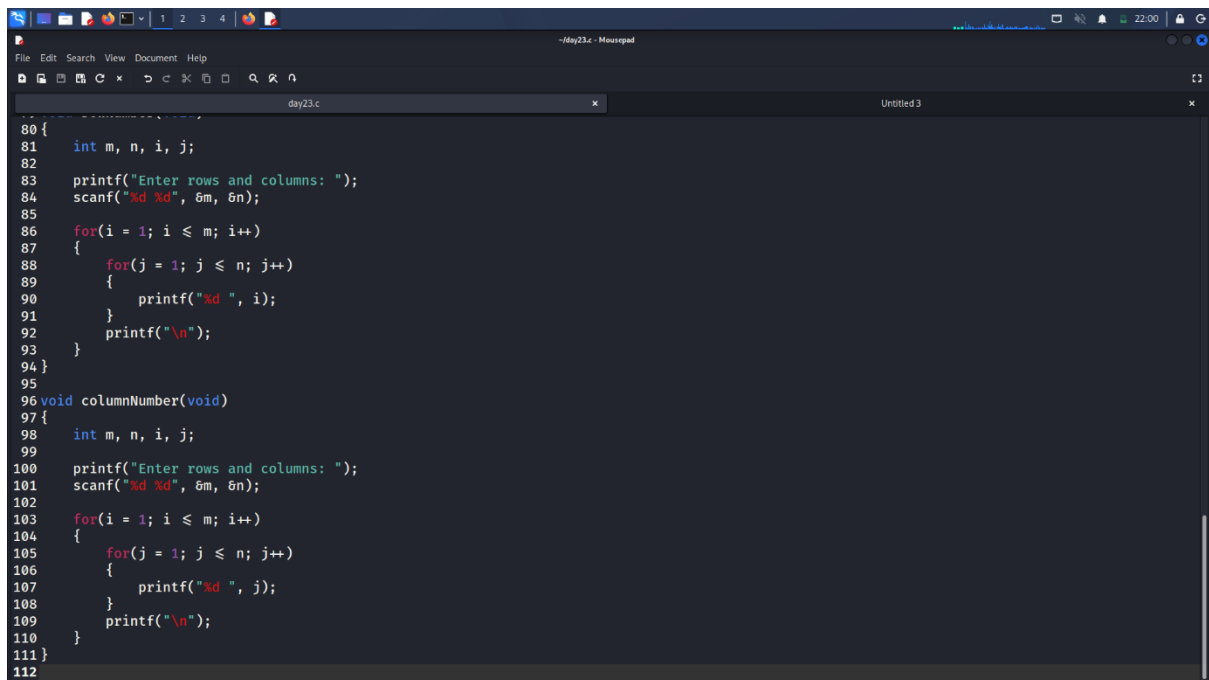


```
1#include <stdio.h>
2
3void solidSquare(void);
4void solidRectangle(void);
5void rowNumber(void);
6void columnNumber(void);
7
8int main()
9{
10    int choice;
11
12    printf("1. Solid Square Pattern\n");
13    printf("2. Solid Rectangular Pattern\n");
14    printf("3. Row-Number Pattern\n");
15    printf("4. Column-Number Pattern\n");
16
17    printf("Enter your choice: ");
18    scanf("%d", &choice);
19
20    switch(choice)
21    {
22        case 1:
23            solidSquare();
24            break;
25
26        case 2:
27            solidRectangle();
28            break;
29
30        case 3:
31            rowNumber();
32            break;
33
34        case 4:
35            columnNumber();
36            break;
37
38        default:
39            printf("Invalid choice\n");
40    }
41}
```



The screenshot shows a code editor window titled "day23.c" with a dark theme. The code defines two functions: `solidSquare` and `solidRectangle`. `solidSquare` takes no arguments and prompts the user to enter the number of rows, then prints a solid square of asterisks. `solidRectangle` takes no arguments and prompts the user to enter rows and columns, then prints a solid rectangle of asterisks. The code is as follows:

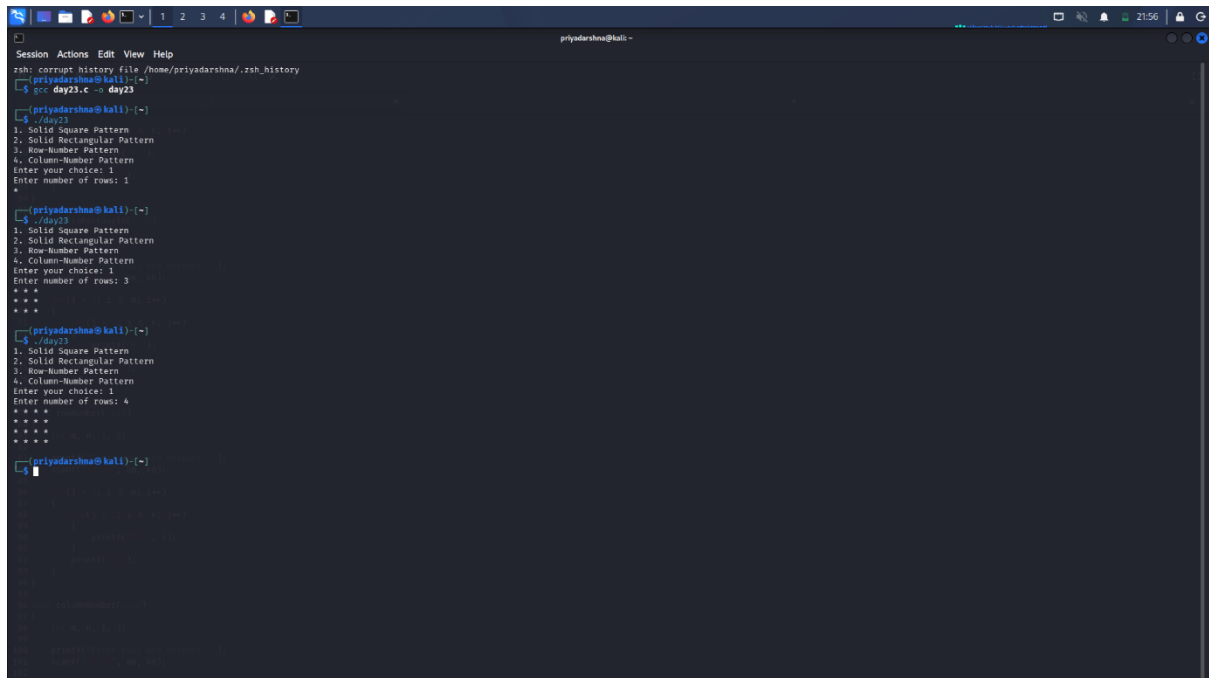
```
42     return 0;
43 }
44
45 void solidSquare(void)
46 {
47     int n, i, j;
48
49     printf("Enter number of rows: ");
50     scanf("%d", &n);
51
52     for(i = 1; i <= n; i++)
53     {
54         for(j = 1; j <= n; j++)
55         {
56             printf("* ");
57         }
58         printf("\n");
59     }
60 }
61
62 void solidRectangle(void)
63 {
64     int m, n, i, j;
65
66     printf("Enter rows and columns: ");
67     scanf("%d %d", &m, &n);
68
69     for(i = 1; i <= m; i++)
70     {
71         for(j = 1; j <= n; j++)
72         {
73             printf("* ");
74         }
75         printf("\n");
76     }
77 }
78
79 void rowNumber(void)
80 {
81     int m, n, i, j;
82 }
```



The screenshot shows the same code editor window, now displaying the `rowNumber` and `columnNumber` functions. `rowNumber` prompts the user to enter rows and columns, then prints the row numbers. `columnNumber` prompts the user to enter rows and columns, then prints the column numbers. The code is as follows:

```
80 {
81     int m, n, i, j;
82
83     printf("Enter rows and columns: ");
84     scanf("%d %d", &m, &n);
85
86     for(i = 1; i <= m; i++)
87     {
88         for(j = 1; j <= n; j++)
89         {
90             printf("%d ", i);
91         }
92         printf("\n");
93     }
94 }
95
96 void columnNumber(void)
97 {
98     int m, n, i, j;
99
100    printf("Enter rows and columns: ");
101    scanf("%d %d", &m, &n);
102
103    for(i = 1; i <= m; i++)
104    {
105        for(j = 1; j <= n; j++)
106        {
107            printf("%d ", j);
108        }
109        printf("\n");
110    }
111 }
112 }
```

>TEST CASE 1 (Table 1 complete 1,2,3 cases): Solid Square Pattern



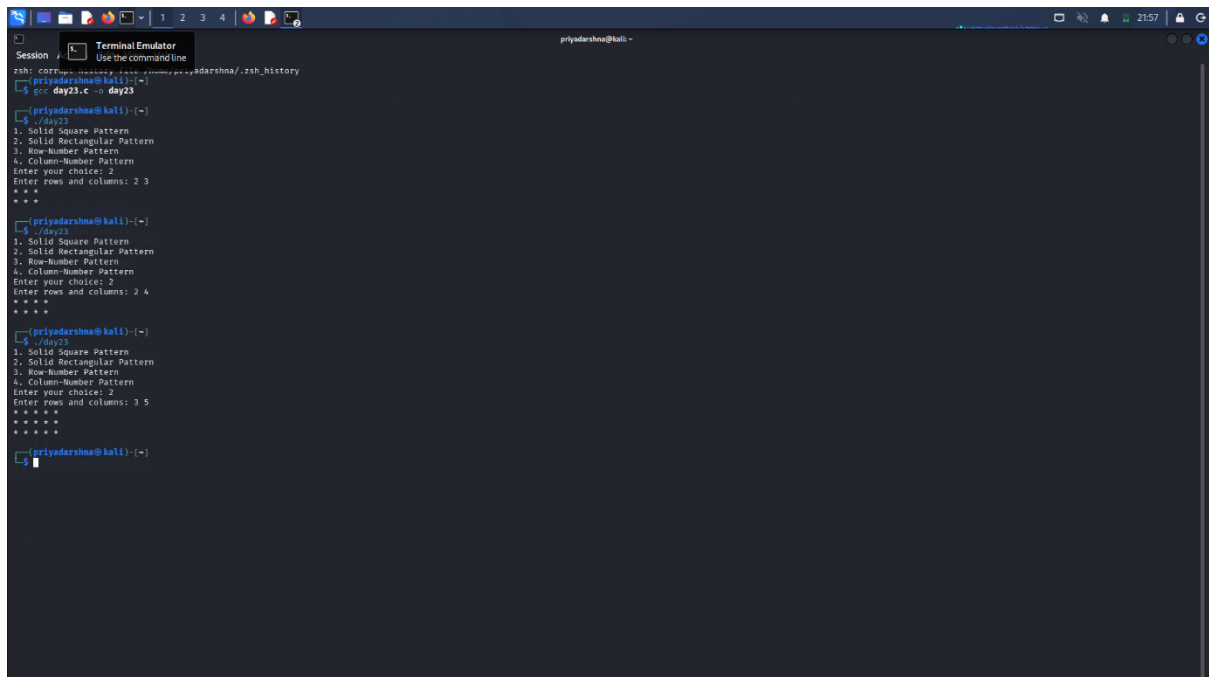
```
Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
[priyadarshna@kali:~]$ gcc day23.c -o day23
[priyadarshna@kali:~]$ ./day23
1. Solid Square Pattern
2. Solid Rectangular Pattern
3. Row-Number Pattern
4. Column-Number Pattern
Enter your choice: 1
Enter number of rows: 1
*

[priyadarshna@kali:~]$ ./day23
1. Solid Square Pattern
2. Solid Rectangular Pattern
3. Row-Number Pattern
4. Column-Number Pattern
Enter your choice: 1
Enter number of rows: 3
***

[priyadarshna@kali:~]$ ./day23
1. Solid Square Pattern
2. Solid Rectangular Pattern
3. Row-Number Pattern
4. Column-Number Pattern
Enter your choice: 1
Enter number of rows: 4
****

[priyadarshna@kali:~]$
```

>TEST CASE 2 (Table 2 complete 1,2,3 cases): Solid Rectangular Pattern



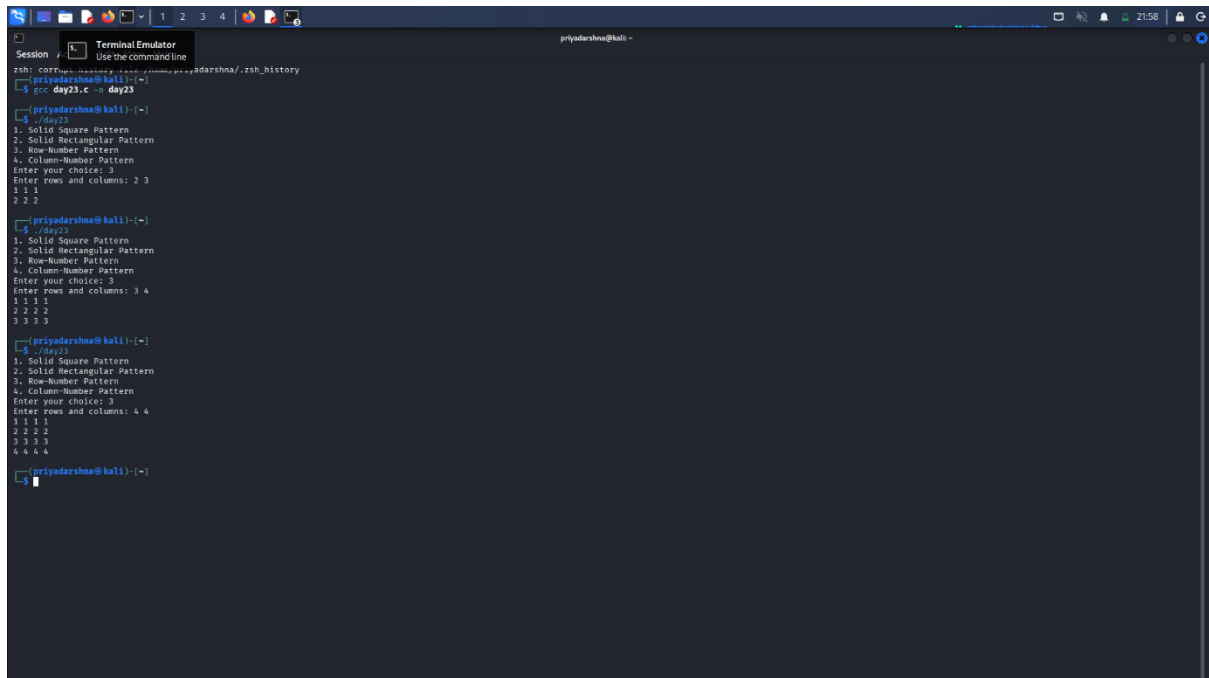
```
Session Terminal Emulator Use the command line
zsh: corrupt history file /home/priyadarshna/.zsh_history
[priyadarshna@kali:~]$ gcc day23.c -o day23
[priyadarshna@kali:~]$ ./day23
1. Solid Square Pattern
2. Solid Rectangular Pattern
3. Row-Number Pattern
4. Column-Number Pattern
Enter your choice: 2
Enter rows and columns: 2 3
***

[priyadarshna@kali:~]$ ./day23
1. Solid Square Pattern
2. Solid Rectangular Pattern
3. Row-Number Pattern
4. Column-Number Pattern
Enter your choice: 2
Enter rows and columns: 2 4
****

[priyadarshna@kali:~]$ ./day23
1. Solid Square Pattern
2. Solid Rectangular Pattern
3. Row-Number Pattern
4. Column-Number Pattern
Enter your choice: 2
Enter rows and columns: 3 5
*****

[priyadarshna@kali:~]$
```

>TEST CASE -3 (Table 3 complete 1,2,3 cases) : Row-number Pattern



```
Session / Terminal Emulator
Use the command line

zsh: corT... history -r... priyadarshna/.zsh_history
$ gcc day23.c -o day23

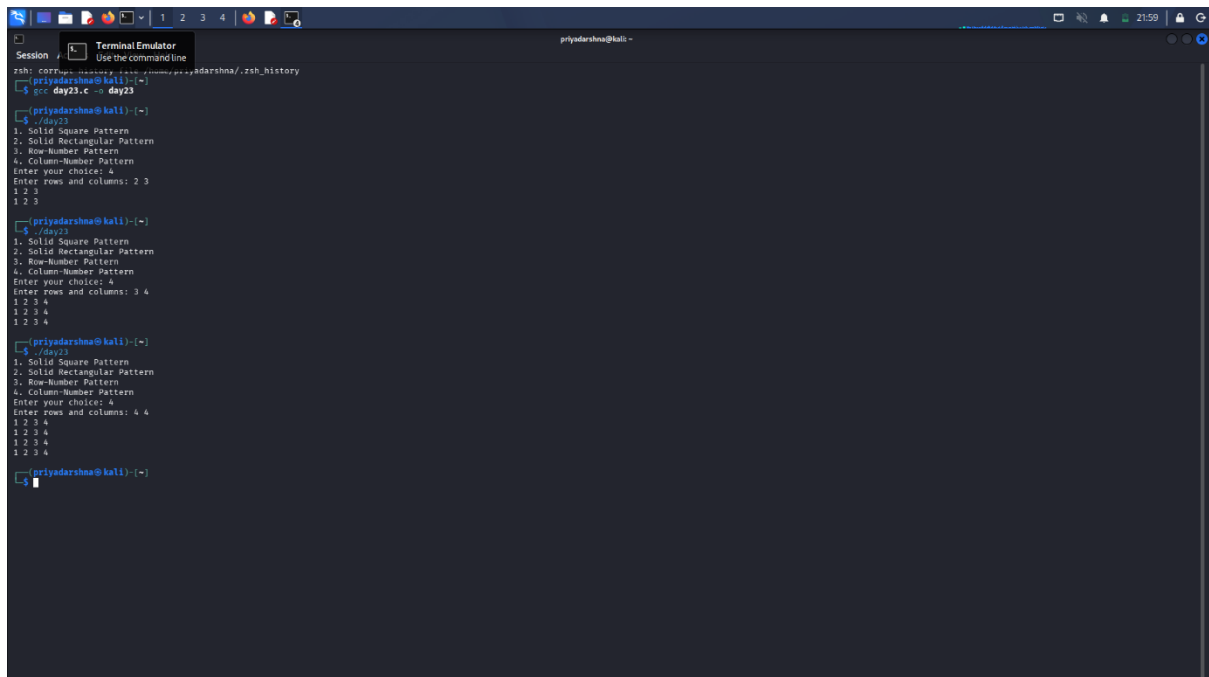
[priyadarshna@kali:~]$ ./day23
1. Solid Square Pattern
2. Solid Rectangular Pattern
3. Row-Number Pattern
4. Column-Number Pattern
Enter your choice: 3
Enter rows and columns: 2 3
1 1 1
2 2 2

[priyadarshna@kali:~]$ ./day23
1. Solid Square Pattern
2. Solid Rectangular Pattern
3. Row-Number Pattern
4. Column-Number Pattern
Enter your choice: 3
Enter rows and columns: 3 4
1 1 1 1
2 2 2 2
3 3 3 3

[priyadarshna@kali:~]$ ./day23
1. Solid Square Pattern
2. Solid Rectangular Pattern
3. Row-Number Pattern
4. Column-Number Pattern
Enter your choice: 3
Enter rows and columns: 4 4
1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4

[priyadarshna@kali:~]$
```

>TEST CASE – 4 (Table 4 complete 1,2,3 cases) : Column-number Pattern



```
Session / Terminal Emulator
Use the command line

zsh: corT... history -r... priyadarshna/.zsh_history
$ gcc day23.c -o day23

[priyadarshna@kali:~]$ ./day23
1. Solid Square Pattern
2. Solid Rectangular Pattern
3. Row-Number Pattern
4. Column-Number Pattern
Enter your choice: 4
Enter rows and columns: 2 3
1 2 3
1 2 3

[priyadarshna@kali:~]$ ./day23
1. Solid Square Pattern
2. Solid Rectangular Pattern
3. Row-Number Pattern
4. Column-Number Pattern
Enter your choice: 4
Enter rows and columns: 3 4
1 2 3 4
1 2 3 4
1 2 3 4

[priyadarshna@kali:~]$ ./day23
1. Solid Square Pattern
2. Solid Rectangular Pattern
3. Row-Number Pattern
4. Column-Number Pattern
Enter your choice: 4
Enter rows and columns: 4 4
1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4

[priyadarshna@kali:~]$
```